Fundamental Thm of convergence theory (Cheeger)

Let $m \geq 2$, $Q > 0$, $m \geq 0$, $\alpha \in (0,1)$, $r > 0$.

Let $\mathcal{M}^{m,\alpha}(n, Q, r)$ = class of (M, g, p) s.t. $\|(M,g)\|_{C^{m,\alpha},r} \leq Q$. Then $\mathcal{M}^{m,\alpha}(n, Q, r)$ is compact in pointed $C^{m,\beta}$-topology for $\beta < \alpha$.

<u>Proof</u> Write $\mathcal{M} = \mathcal{M}^{m,\alpha}(n, Q, r)$

<u>Step 1</u> <u>Claim</u> $\mathcal{M}$ is pre-compact in pointed GH-topology.

Need to show that given $\varepsilon, R > 0$ $\exists\, C(\varepsilon, R)$ s.t. for any $(M, g, p) \in \mathcal{M}$ the ball $B(p, R)$ contains at most $C(\varepsilon, R)$ disjoint ε-balls.

Let $B(x_1, \varepsilon), \dots, B(x_N, \varepsilon)$ be disjoint

If ε is small enough, then any $B(x_i, \varepsilon)$ is contained in some chart

$$\varphi : B(0, r) \longrightarrow U \subset M$$

$$\Longrightarrow \quad \varphi^{-1}(B(x_i, \varepsilon)) \supset B(\varphi^{-1}(x_i), e^{-Q}\varepsilon)$$

$$\Longrightarrow \quad \mathrm{Vol}(B(x_i, \varepsilon)) \geq c_1 \mathrm{Vol}(B(0, \varepsilon))$$

We need an upper bound for $Vol(B(p,R))$:

1. $\exists\ \delta, K > 0$ s.t. every δ-ball $B(x,\delta)$ can be covered by K $\delta/4$-balls (Work in charts)

2. By induction, every ball $B(x, \ell\,\delta/2)$ can be covered by K^ℓ $\delta/4$-balls: $\ell = 1$ follows from 1.

   <u>induction step</u>: if $B(x, \ell\,\delta/2)$ is covered by $B(y_1, \delta/4), \dots, B(y_{K^\ell}, \delta/4)$

   then $B(x, (\ell+1)\delta/2) = B(x, \ell\,\delta/2 + \delta/2)$ is covered by $B(y_i, \delta)$ $i = 1, \dots, K^\ell$

   But each $B(y_i, \delta)$ is covered by K $\delta/4$-balls

   $\Rightarrow B(x, (\ell+1)\delta/2)$ is covered by $K \cdot K^\ell = K^{\ell+1}$ $\delta/4$-balls.

3. If δ is very small, then

   $Vol(B(x;\delta)) \le c_2 \delta^n$

   $\Rightarrow Vol(B(p, \ell\,\delta/2)) \le c_2 (\delta/4)^n \cdot K^\ell$

Lower bound for $Vol(B(x_i, \varepsilon))$

+ upper bound for $Vol(B(p,R))$

$\Rightarrow$ estimate for number of disjoint $B(x_i, \varepsilon) \subset B(p,R)$

Gromov $\Rightarrow$ choosing a subsequence we can assume $(M_i, g_i, p_i) \longrightarrow (X, d_X, p)$ in pointed GH-sense

metric space.

<u>Step 2</u> <u>Claim</u>: (X, d_X, p) is isometric to a Riemannian mfld (M, g, p) with $\|(M,g)\|_{C^{m,\alpha}, r} \leq Q$.

Let $q \in X$, $q_i \to q$ with $q_i \in M_i$.

We have charts $\varphi_i : B(0,r) \to U_i \subset M_i$
$\varphi_i(0) = q_i$

The φ_i are uniformly Lipschitz (using $|D\varphi_i| \leq e^Q$)

$\Rightarrow$ up to choosing a subsequence φ_i converge to $\varphi : B(0,r) \to X$
$\varphi(0) = q$

$$e^{-Q} \min\{|x_1 - x_2|, 2r - |x_1| - |x_2|\} \leq d_X(\varphi(x_1), \varphi(x_2)) \leq e^Q |x_1 - x_2|$$

$\Rightarrow$ φ is a homeomorphism onto its image

$\Rightarrow$ X is a topological manifold.

X is covered by charts $\varphi_s : B(0,r) \to X$ obtained as limits of $\varphi_{is} : B(0,r) \to M_i$ as $i \to \infty$

On each such chart we have metrics $\varphi_{is}^* g_i \xrightarrow{C^{m,\beta}} g_s$ on $B(0,r)$

$g_s \in C^{m,\alpha}$

satisfying $e^{-2Q}\delta_{jk} \le g_{s,jk} \le e^{2Q}\delta_{jk}$

$$r^{|I|+\alpha}\,\|\partial^I g_{s,jk}\|_{C^\alpha} \le Q$$

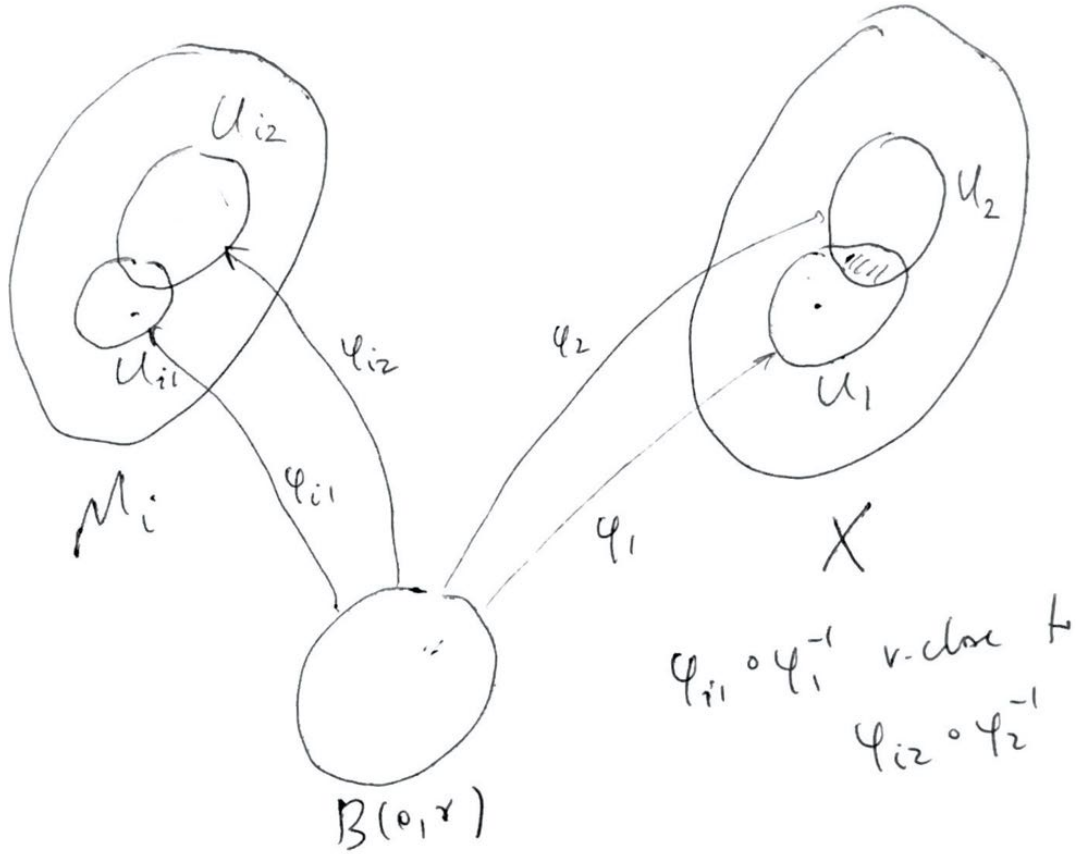
The limit g_s defines the same distance (locally) on $\varphi_s(B(0,r)) \subset X$ as d_X.

<u>Note</u>: At this point the transition maps $F_{st} = \varphi_s^{-1}\circ\varphi_t$ are only Lipschitz

but $F_{st} : (V, g_t) \longrightarrow (F_{st}(V), g_s)$

$V \subset B(0,r)$, $F_{st}(V) \subset B(0,r)$

is an isometry. Calabi - Hartman $\Rightarrow F_{st} \in C^1$.

$\Rightarrow$ X is a differentiable manifold with $C^{m,\alpha}$ Riemannian structure (X, g)

<u>Step 3</u> <u>Claim</u>: $(M_i, g_i, p_i) \longrightarrow (X, g, p)$ in $C^{m,\beta}$.